Humanities and Social Sciences Communications

Article in Press

https://doi.org/10.1057/s41599-026-07998-z

Financial development and renewable energy consumption: a novel view under the moderating role of institutional quality across APEC economies

Received: 1 May 2025

Accepted: 12 June 2026

Cite this article as: Chowdhury, M.A., An, M., Hedvicakova, M. Financial development and renewable energy consumption: a novel view under the moderating role of institutional quality across APEC economies. *Humanit Soc Sci Commun* (2026). https://doi.org/10.1057/s41599-026-07998-z

Mohammad Abir Shahid Chowdhury, Min An & Martina Hedvicakova

Financial development and renewable energy consumption: a novel view under the moderating role of institutional quality across APEC economies

Abstract: Sustainable development through economic growth has led to the importance of transitioning to renewable energy. However, APEC's heavy reliance on fossil fuels and various levels of economic growth generate unique challenges in attaining these sustainability goals. While existing literature focuses on financial development (FD) influencing energy transitions, it overlooks how institutional quality (IQ) influences the effectiveness of FD in promoting renewable energy consumption (REC). This study analyzes 12 APEC economies from 2002 to 2021 to explore the dynamics of FD, REC, and the moderating role of IQ. The study applied fixed-effects and panel quantile regression, and robustness checks by varying explanatory variables and replacing econometric methodologies. The findings reveal that FD has a negative effect on REC, mainly at the median quantile, indicating a bias towards traditional energy investments. IQ positively moderates the FD–REC relationship but has no direct significant impact on REC. Moreover, energy intensity (EI) and technological innovation (TI) negatively enhance REC, whereas trade openness (TO) has a positive but insignificant association with REC. Finally, this study has policy implications for APEC economies to strengthen institutional frameworks, boost REC, and contribute to global climate targets, advancing SDG 7 (affordable and clean energy) and SDG 13 (climate action).

Keywords: Financial development, renewable energy consumption, institutional quality, sustainable development goals, APEC.

1. Introduction

The escalating threat of climate change has made the world focus on sustainable development, and renewable energy is leading the way in this endeavor. The energy transition is not just a technological shift but a response to one of the most pressing global issues reducing carbon emissions, mitigating environmental degradation, and meeting the growing energy demands of the world's population. As such, renewable energy consumption (REC) lies at the heart of this transition, directly supporting SDGs 7 (affordable and clean energy) and 13 (climate action). The global commitment to sustainability is further emphasized by the agreements made during the COP28 and COP29 conferences, where nations renewed their pledges to accelerate the transition to clean energy and align their energy policies with the SDGs (Behera, Sucharita, et al., 2024; Clémençon, 2024). The twenty-one member nations of the Asia-Pacific Economic Cooperation (APEC) economies consume more than fifty percent of global energy demand, making it a crucial region for climate action and the development of renewable energy. This region presents both a significant challenge and an opportunity for accomplishing the SDGs because of its high energy demand and diverse economic systems. Despite the growing momentum for renewable energy adoption, APEC economies face unique challenges because of their heavy reliance on fossil fuels, which complicates efforts to transition to more sustainable energy systems. In this context, the role of financial development in fostering REC is more crucial than ever.

Studies have explored whether financial development (FD) is a central element in the renewable energy transition, as efficient financial systems can provide capital flows that support investments in green technologies and clean energy projects. This most often refers to the ability of financial markets and institutions to efficiently allocate resources (Behera, Sucharita, et al., 2024). The theoretical framework posits that a robust financial system may facilitate rapid renewable energy transitions by offering reasonable posts to green technologies, funding energy infrastructure and attracting investments towards it (G. Sun et al., 2023). However, empirical studies found that FD may not always lead to faster adoption of renewable energy with existing risk perceptions, economic structure, and lack of initiative to motivate sustainable energy activities result in more of financial systems favoring the traditional energy investment (Alshagri et al., 2024; Li, 2023). This inconsistency suggests that financial development may not be sufficient to catalyze the energy transition, a gap that has spurred scholars to examine the role of institutional quality (IQ).

The importance of integrating IQ into renewable energy transitions is crucial, as governance structures can either facilitate or hinder financial flows into the adoption of green technologies and clean energy projects. Studies by Ahmad et al. (2023) and Zhang et al. (2023) imply that strong institutional quality can boost the financial development effectiveness of renewable energy by dampening credit risk, promoting transparency, and optimizing resource utilization. Conversely, poor IQ may lead to corruption, lack of accountability, and obtrusive policies that form barriers to investors in renewable energy (Gani, 2021). The United Nations Sustainable Development Goals (SDGs), particularly SDG 7 and SDG 13, are also linked to the role of institutions in promoting renewable energy adoption (Behera, Behera, et al., 2024; Mustafa et al., 2022; Ofori et al., 2023). Governments can seek to improve institutional quality to build the framework needed to attract long-term investments in sustainable energy, thus aiding SDG 7 on affordable and clean energy (Huck, 2023). Moreover, such high-quality institutions also make it easier to adhere to environmental, social, and governance (ESG) standards, which ultimately encourage ethical investment practices allowing for a clean energy transition to attain SDG 13 (Riti et al., 2021). Therefore, governments in the APEC region must prioritize on enhancing institutional quality to bring off the full benefits that financial development can have on fostering renewable energy growth.

Despite widespread acknowledgment of the role of financial development in promoting renewable energy expansion, the existing literature falls short of demonstrating how IQ affects this relationship, especially in the context of APEC economies. Several studies primarily focus on how FD influences energy transitions Liu et al. (2023) and Saadaoui & Chtourou (2023); however, the influence of IQ on the effectiveness of financial resources in promoting renewable energy projects remains unexplored. To probe this gap, the current study explores the moderating role of IQ in the link between FD and REC in APEC countries. Furthermore, existing studies often rely on mean-based estimators, obscuring important distributional differences in the FD-REC link, particularly in APEC, where institutional and financial conditions vary across countries. This study also addresses this gap by employing quantile regression to capture heterogeneous effects, offering more nuanced insights into how FD influences REC at different stages of the energy transition, tailored to APEC's diverse institutional contexts.

The primary objectives of this study are multifaceted. This study first investigates the exact extent to which FD directly affects REC in APEC countries, including the impact of financial availability on clean energy investments. Second, it explores how IQ moderates the FD–REC nexus and highlights the importance of institutional frameworks in enhancing FD's role in sustainable energy transitions. Finally, this research endeavors to bestow on the ongoing discourse on sustainable development by delivering empirical support on the moderating effects of IQ on renewable energy adoption in upper income and upper-middle income economies.

This study makes several key contributions to the extant literature on renewable energy and sustainable development. It extends the understanding of the FD–REC nexus by incorporating IQ as a moderating variable, offering a more comprehensive perspective on how institutional frameworks influence the relationship between financial systems and sustainable energy transitions. The research also provides practical implications for policymakers, financial institutions, and environmental advocates. Policymakers in APEC economies can use these findings to design strategies that enhance institutional quality, thereby maximizing the impact of financial development on renewable energy adoption. Financial institutions can leverage the insights from this research to refine their investment policies, focusing on renewable energy projects in regions with strong institutional support. Additionally, the study applies advanced econometric methods to ensure rigorous and reliable conclusions, making it a valuable addition to the field. Finally, the research emphasizes the significance of this region in global energy markets and sustainability efforts by focusing on APEC economies. Hence, this article contributes to ongoing efforts to achieve SDGs 7 and 13 by exploring the relationship between financial development, institutional quality, and renewable energy consumption in the APEC region. Given the magnitude of APEC economies in the global energy transition, the findings of this research offer timely recommendations for policymakers, financial institutions, and stakeholders seeking to accelerate clean energy transitions and contribute to global sustainability goals.

The empirical research is designed as follows: Section 2 delves into the critical review of the literature. Section 3 uncovers the dataset, model and methodology. Section 4 presents the econometric outcomes and discusses the results. Finally, Section 5 summarizes the conclusions, policy implications, and limitations, as well as future directions.

2. Literature Review

Although a substantial amount of literature exists on the FD and REC nexus in developed economies, no clear consensus has been drawn, and the associations between FD and REC remain opaque in the APEC region.

2.1 Financial Development (FD) and Renewable Energy Consumption (REC) nexus

Existing studies have shown varying outcomes of FD and REC linkages across various regimes. For instance, Shahbaz et al. (2021) examined the correlation between FD, economic growth, consumer prices, and REC in 34 upper-middle-income countries. The study identified a positive interplay between FD and REC. The authors further recommended that investors and policymakers build strategic policy frameworks to maximize FD's role in enhancing REC. Similarly, Z. Sun et al. (2023) applied two-step GMM method on developed and developing economies to investigate the effect of FD on REC. Their results revealed a statistically positive association between these two variables. The authors further recommended that investors and policymakers build strategic policy frameworks to maximize FD's role in enhancing REC. Again, Y. Wang et al. (2024) applied social-ecological interaction theories and the EKC framework over the period of 2001 to 2020 in OECD countries. They reported that technological innovation and social sustainability curb carbon emission, with targeted economic growth drive decarbonization in emerging economies. Hasanov et al. (2021) constructed a theoretical framework for the time span 1990-2017 in BRICS economies, and found that technological progress and renewable energy reduce emissions, while trade openness increases pollution. The outcome emphasize that financial development must align with REC policies to foster sustainable energy transitions. Xu et al. (2022) also applied the EKC hypothesis in E7 countries for the period of 1992 to 2020 and explored that renewable energy structure reduces ecological footprints, but natural resource rents and policy uncertainty hinder environmental progress. The study emphasized that economic policy is crucial to mitigate volatility and reform resource governance for REC investment. Conversely, several studies revealed unresponsive or ambiguous outcomes regarding the influence of FD on clean energy usage. Yue et al. (2019) conducted a comparable study on 21 transitional countries by applying the PSTR model. The research revealed no significant linear relationship between variables, aligned with the outcome of (Abbasi & Riaz, 2016; Shahbaz et al., 2016).

Nevertheless, prior studies have noted a negative association between FD and REC in APEC countries. For instance, J. Wang et al. (2021) explored the influence of FD on REC in China from 1997 to 2017, finding that FD influenced the REC negatively in the long run. The author argued that low-quality financial institutions and higher financial costs prevent the investments in renewable energy infrastructures and technologies. This outcome is also supported by (Ouyang & Li, 2018). Khan et al. (2020) also employed Chinese data from 1987 to 2017, and found that natural resource abundance negatively impacts FD, while technological innovation, trade openness, and human capital positively drive financial sector growth. This highlights that FD's capacity to fund REC is contingent on overcoming resource dependence and investing in innovation and human capital. Again, Farhani & Solarin (2017) explored the long-run relationship between FD, economic growth, FDI, trade, capital, and renewable energy demand in the U.S. The study utilized the structural break and Bayer–Hanck cointegration approach and time series data over 1973q1–2014q4 and found that FD incurs renewable energy demand in the US economy. The study suggests that FD, FDI, and trade decline energy demand due to limited renewable energy infrastructure. Similarly, Danish et al. (2018) investigated the impact of financial development and globalization on energy consumption in Next-11 countries from 1990 to 2014 using a panel estimation technique. The author discovered that financial development effectively declined clean energy consumption in Mexico and Korea. The study confirms bidirectional causality between financial growth and energy consumption, and emphasizes the need for policymakers in emerging economies to implement energy conservation strategies and manage the impact of globalization on green energy use. So based on above arguments, the study stated that FD may inhibit REC and slow down the sustainable energy transition in APEC nations. To investigate the relationship posited in the literature review, this study tests the following hypotheses:

H_1: Financial development has a significantly negative relationship with renewable energy consumption in APEC nations.

2.2 Institutional Quality (IQ) and Renewable Energy Consumption (REC) Nexus

The relationship between IQ and REC has received limited attention in the literature, with several studies highlighting mixed findings. While some studies suggest that weak governance and ineffective policy frameworks hinder renewable energy development, others emphasize the adverse effects of institutional frameworks and institutional inefficiencies on sustainable energy transitions. For instance, Saba & Biyase (2022) examined the potential factors that influence REC in 35 European countries between 2000 and 2018. This study employed a system-GMM approach and argued that IQ has a statistically insignificant impact on renewable energy development. The study suggests that ineffective governance and policy frameworks hinder renewable energy development. Likewise, Acheampong et al. (2021) explored the causal relationships between institutions, renewable energy, carbon emissions, and economic growth for the time span 1960–2017 in 45 sub-Saharan African countries. They found no direct link between IQ and economic growth. The study concluded that weak institutional frameworks may hinder effective policy implementation, affecting the relationship between IQ, REC, and carbon emissions. Arminen & Menegaki (2019) investigated the causal relationships between economic growth, energy consumption, corruption, and CO2 emissions in high- and upper-middle-income countries over the period 1985-2011. The findings revealed that IQ, particularly corruption, exerts a limited influence on energy and environmental policies relative to climate factors. These outcomes imply that IQ plays a limited role in influencing energy consumption and environmental outcomes in these regions. Apergis & Pinar (2021) also showed that institutional quality, measured by the effects of party polarization, is inversely related to REC in 25 European Union countries from 2003 to 2017. The results highlight that increased political polarization hampers REC owing to its detrimental impact on policy coherence and the effectiveness of institutions in advancing sustainable energy transitions. This outcome is also supported by (Potrafke, 2010; Wen et al., 2016). Again, E. Wang et al. (2022) explored the influence of political risk on REC for 32 OECD countries from 1997 to 2019, finding that political risk inversely influenced the REC. Moreover, Islam et al. (2022) explored the role of renewable and non-renewable energy in sustainable development, along with FD and IQ. They reported that IQ and FD adversely affect sustainable development. They further added that the negative association of IQ is attributed to its insufficient role in fostering effective resource management and technology diffusion for sustainable development. Following the above discussion, we can infer that institutional quality has an adverse effect on REC in the APEC region.

2.3 Financial development (FD), Institutional Quality (IQ) and Renewable Energy Consumption (REC) Nexus

The interplay between FD and REC has been widely explored, with an increasing emphasis on the moderating role of IQ. IQ may enhance the impact of FD on REC by strengthening institutional frameworks and supporting sustainable energy transitions. For instance, Hussain & Dogan (2021) employed second-generation econometric methods within an EKC framework for the time span of 1992 to 2016 in BRICS to find that environment-related technologies and IQ reduce ecological footprints while the EKC hypothesis is unsupported. This outcome highlights that strong institutions and green technology investments are prerequisites for integrating REC into sustainable development goals. Behera, Behera, et al. (2024) applied the method of moments quantile regression for 14 emerging economies over the period 1990 to 2021. They found that renewable energy and green finance reduce emissions, whereas political stability alone increases them. However, they explored the moderating effect of political stability and green finance on emissions, emphasizing that stronger political systems are crucial for fostering green finance and renewable energy for SDG attainment. Basheer et al. (2024) examined the correlation among FD, globalization, energy consumption, and corruption control in South Asian economies for the time span 1996-2019. The findings reveal that energy consumption, FD, and globalization negatively affect environmental quality, but corruption control measures can mitigate these adverse impacts. The author suggests that anti-corruption initiatives are crucial for promoting sustainable development. Again, Mesagan & Olunkwa (2022) applied PMG approach to 18 African nations between 1996 and 2017 to investigate the moderating role of regulatory quality on energy use and FD in pollution reduction. The study discovered that effective regulatory quality enhances the capacity of energy use and FD to mitigate environmental pollution. Building on the above discussion, we can infer that IQ acts as a moderating mechanism, amplifying the positive effects of FD on REC. This leads to the following testable hypothesis:

H_2: Institutional quality significantly positively moderates the relationship between financial development and renewable energy consumption.

Research Gap

Despite considerable research on the relationship between financial development (FD) and renewable energy consumption (REC), significant gaps remain in our understanding of this dynamic in the APEC region. Existing studies predominantly focus on developed economies, leaving the unique institutional and economic contexts of APEC nations underexplored. Given APEC's diverse institutional landscapes, the region presents a unique opportunity to investigate how varying levels of IQ influence the relationship between FD and REC. Such an analysis is crucial for informing policies in both emerging and developed economies within APEC. Moreover, prior research has primarily relied on mean-based panel estimators, which offer an average effect and obscure important distributional heterogeneity in the relationship between foreign direct investment (FDI) and REC. This is particularly problematic in APEC, where institutional constraints and financing friction vary across countries. The marginal effect of FD on REC may differ significantly across economies, depending on their levels of renewable energy adoption or financial development. To address this gap, this study employs quantile regression to uncover the heterogeneous effects that are often masked by average estimates. This approach allows for a deeper understanding of how FD influences REC at different stages of the energy transition, providing policy-relevant insights tailored to the APEC region at varying points in their renewable energy transitions.

3. Theoretical Underpinning, Dataset, Model and Methodology

3.1. Theoretical Underpinning

In the pursuit of enhancing environmental sustainability in APEC, two core theoretical strands underpin this study: energy transition theory (ETT) and institutional theory (IT). ETT suggests that energy transitions are complex, multidimensional processes driven by technological innovation, sociopolitical changes, and market dynamics (Geels, 2002; Geels et al., 2017). This theory emphasizes that the shift from conventional energy sources to renewable energy is not merely a technical change but also involves broader societal, economic, and institutional structures. Geels also highlighted that technological innovation, such as the adoption of renewable energy, is influenced by economic forces, policy frameworks, and societal readiness. In the context of the FD and REC nexus, ETT underscores that the availability of financial resources is critical for the adoption of clean energy technologies (Yadav et al., 2024). However, the success of these technologies also depends on the broader institutional environment, as explored by IT.

IT posits that organizational behavior is shaped by formal and informal rules within an institutional environment (North, 2005). IQ encompasses governance aspects, such as government effectiveness, regulatory quality, rule of law, and political stability, which play pivotal roles in determining the efficiency of energy transitions (Amenta & Ramsey, 2010). Strong institutions are critical for reducing transaction costs, mitigating risks, and providing the regulatory certainty necessary to attract financial investment in renewable energy projects (Polzin et al., 2017). Thus, IQ is hypothesized to moderate the relationship between FD and REC, aligning with Zhang et al. (2023), who concluded that institutional frameworks can either facilitate or hinder the flow of financial resources into sustainable energy projects. In contrast, weaker institutional environments often lack the regulatory stability, governance capacity, and policy coherence necessary to attract and efficiently allocate financial resources, thereby hindering the effective deployment of renewable energy projects (Makhdum et al., 2022).
Therefore, the integration of energy transition theory and institutional theory in this study provides a robust framework for understanding how FD and IQ interact to influence REC, offering valuable insights for policy development and energy transitions within the diverse institutional contexts of APEC economies.

3.2 Data

Data from various sources were employed to assess the role of the interaction variable IQ on the FD and REC nexus and to examine the impact of the control variables energy intensity (EI), trade openness (TO), and technological innovation (TI) on REC (see Table 1). The data were collected from the World Bank (WB) database. The dataset encompasses the APEC region, limited to 12 countries (Australia, China, Hong Kong SAR, Japan, Republic of Korea, Malaysia, Mexico, Philippines, Singapore, Thailand, the United States, and Vietnam) owing to data unavailability and mismatching between data sources in this analysis for the period 2002–2021. This study utilized principal component analysis (PCA) to develop an institutional quality index, employing a set of six government indicators from the World Bank. The selected indicators are “Control of Corruption,” “Government Effectiveness,” “Political Stability and Absence of Violence/Terrorism,” “Regulatory Quality,” “Rule of Law,” and “Voice and Accountability.” As REC data are available in the WB until 2021 and institutional quality index variable data have been available since 2002, the time frame is limited to 2002–2021. This study acknowledges the interaction between financial development and the institutional quality index as innovation. Table 1 presents the key parameters and data sources.

Table 1 Descriptions of variables

Variables	Signs	Measurement	Source
Renewable Energy Consumption	REC	Renewable energy consumption (% of total final energy consumption)	World Development Indicators
Financial Development	FD1	Domestic credit to the private sector by banks (% of GDP)	World Development Indicators
Financial Development	FD2	Stocks Traded % of GDP	World Development Indicators
Institutional Quality	IQ	IQ index prepared by Principal component analysis (PCA) method using six government indicators (Control of Corruption: Estimate; Government Effectiveness: Estimate; Political Stability and Absence of Violence/Terrorism: Estimate; Regulatory Quality: Estimate; Rule of Law: Estimate; and Voice and Accountability: Estimate)	Author's Calculation
Energy Intensity	EI	Energy intensity level of primary energy (MJ/$2021 PPP GDP)	World Development Indicators
Trade Openness	TO	TRADE (% OF GDP)	World Development Indicators
Technological Innovation	TI	Patent applications, nonresidents, and residents	World Development Indicators

Note: Table 1 depicts the sign, measurement, and data sources of indicators.

3.3 Model Specifications

The present work employs REC as an explained parameter and FD, EI, TO, and TI as explanatory parameters. It analyzes the dynamic association between REC and FD with FD_1 and FD_2. It also investigates the interaction between IQ and the impact of FD on REC with the independent variable FD*IQ. The econometric models are presented in Eq. (1) and (2).

$$REC_{it} = (FD1_{it}, FD2_{it}, IQ_{it}, EI_{it}, TO_{it}, TI_{it}) \quad (1)$$

$$REC_{it} = f_1(FD1_{it}, FD2_{it}, IQ_{it}, EI_{it}, TO_{it}, TI_{it}, FD1_{it} * IQ_{it}) \quad (2)$$

The logarithms of the variables were employed to scrutinize multicollinearity and endogeneity issues and to ascertain a more unbiased evaluation in Eq. (3).

$$\text{Ln}(REC_{it}) = \beta_0 + \beta_1 \ln(FD1_{it}) + \beta_2 \ln(FD2_{it}) + \beta_3 \ln(IQ_{it}) + \beta_4 \ln(EI_{it}) + \beta_5 \ln(TO_{it}) + \beta_6 \ln(TI_{it}) + \mu_{it} \quad (3)$$

There are limited econometric functions with FD and REC variables in the extant APEC literature. The association between FD and REC in the APEC region is examined in this study. Furthermore, this study extends the impact of FD and REC by employing the IQ interaction parameter, as in Eq. (4).

$$\text{Ln}(REC_{it}) = \beta_0 + \beta_1 \ln(FD1_{it}) + \beta_2 \ln(FD2_{it}) + \beta_3 \ln(IQ_{it}) + \beta_4 \ln(EI_{it}) + \beta_5 \ln(TO_{it}) + \beta_6 \ln(TI_{it}) + \beta_7 \ln(FD1_{it}) * \ln(IQ_{it}) + \mu_{it} \quad (4)$$

In the econometric equations, i, t, and μ_{it} denote cross-sectional units (APEC economies), time periods, and error terms, respectively. The constant term is denoted by β_0, while β_1 to β_7 signify the elasticity coefficients.

3.4 Research Methodology

Diagnostic tests

This study conducted several diagnostic analyses to apply the most accurate and suitable estimation technique to achieve robust outcomes. First, the VIF test was applied following descriptive statistics and a correlation matrix to confirm the absence of multicollinearity and validate the robustness of the latent determinants. Next, the Pesaran (2021) cross-sectional dependence (CSD) test and Hashem Pesaran & Yamagata (2008) heterogenous slope (HS) test were employed to prevent biased parameter estimation, and mitigate the challenges posed by the HS. Following the CSD and HS estimations, this study employed the Levin-Lin-Chu (LLC) Levin et al. (2002) test and the Im-Pesaran-Shin (IPS) Im et al. (2023) tests to examine the stationarity of the variables. The IPS test extends the LLC framework by allowing heterogeneity in the autoregressive coefficient across cross-sections (Junejo et al., 2026). In addition, this study applied the Pedroni (2004) panel cointegration test between explanatory and explained parameters to assess the long-run nexus. The cointegration test ensures a stable equilibrium relationship as the determinants move together over time (Pedroni, 2004). Subsequently, the study applied the Hausman (1978) test to select between fixed-effects (FE) and random-effect (RE) models. Furthermore, this study used the test of Granger causality Granger (1969) to determine the causal relationships and endogeneity issues among these variables, which is crucial for understanding the direction of causality between the parameters.

Panel econometric tests

Following Athari (2024) and Saliba et al. (2023), this study augments the FE approach with the quantile regression approach to explore how different variables are related to each other in different parts of the dependent variable distribution. Baltagi (2005) and Hsiao (2022) argued that panel data estimation technique increases estimation efficiency, reduces omitted variable bias risk, and mitigates multicollinearity and heterogeneity concerns. Therefore, this study relied on the Hausman test result of 23.12***(0.001) when choosing between FE and RE. The FE model was utilized to account for individual-specific effects, ensuring that the estimates reflect within-country variations, and to capture unobserved heterogeneity that might affect the results. The FE approach is based on Eq. (5).

$$Y_{it} = \alpha + \beta_1 X_{it} + \beta_2 Z_{it} + \mu_i + \epsilon_{it} \qquad (5)$$

In contrast, panel quantile regression allows for a more comprehensive analysis by estimating conditional quantile functions, providing insights into the relationship between the determinants and the dependent variable at different points of the conditional distribution, beyond just the mean. Mean regression methods are unlike quantile regression, which is particularly useful for analyzing how explanatory variables determine the response at different quantiles. The objective function of the quantile regression estimator is as follows, which is minimized in Eq. (6).

$$\min_{\beta} \sum_{i=1}^{N} \rho_q (y_i - X_i' \beta) \qquad (6)$$

Robustness tests

To ensure the robustness of the findings, this study further applied panel quantile regression, fixed effects regression, and system GMM by varying the explanatory variables. This approach accounts for potential endogeneity and reverse causality and provides more reliable estimates by using internal instruments for potentially endogenous variables (Khatib, 2025; Mwanjilinji et al., 2025). Moreover, the modified method of quantile regression (MMQR) and Driscoll-Kraay standard errors were applied to address issues such as autocorrelation, heteroscedasticity, and cross-sectional dependence in the residuals, thereby strengthening the validity of the results (Driscoll & Kraay, 1998; Song & Taamouti, 2021; Troster, 2018). Figure 1 presents a research methodology flowchart that depicts the overall empirical methods.

4. Univariate & Multivariate Results

4.1 Univariate Analysis

Table 2 Descriptive Summary

Variable	Observation	Mean	St. dev	Minimum	Maximum
LnREC	240	1.6970	1.5762	-2.3026	3.9551
LnFD	240	4.4470	0.6416	2.5025	5.5575
LnIQ	240	1.4400	0.5494	0.2623	2.0409
LnEI	240	1.4162	0.4250	0.1570	2.3860
LnTO	240	4.3048	0.7443	3.0178	6.0927
LnTI	240	10.2765	1.9149	6.7499	14.2765

Note: Table 2 presents a descriptive summary of the variables.

Table 2 reveals the descriptive statistics for the APEC nations' cluster. As depicted, the mean of LnTI (10.2765) is paramount and ranges from 6.7499 to 14.2765. This is accompanied by LnFD (4.4470), which varies between 2.5025 and 5.5575. Table 2 also reveals that the mean values of LnEI (1.4162) and LnIQ (1.4400) are the smallest. Furthermore, the descriptive findings illustrate that LnTI (1.9149) has the paramount standard deviation, whereas LNEI and LNIQ, with 0.4250 and 0.5494 standard deviations, have the smallest deviations among other determinants.

Table 3 Pearson correlation matrix

	LnFD	LnIQ	LnEI	LnTO	LnTI	VIF
LnFD	1					2.69
LnIQ	0.4160***	1				2.30
LnEI	0.0248	(-0.2373)***	1			1.96
LnTO	0.3046***	(-0.1777)***	(-0.4441)***	1		2.02
LnTI	0.2014***	0.3413***	0.4241***	(-0.6038)***	1	1.90

Note: Table 3 presents the Pearson correlation matrix and variance inflation factor (VIF) among the independent variables. *** denotes significance at 1% level.

Table 3 reveals the significant association among the explanatory variables and different degrees of negative and positive correlations. These findings underline that multicollinearity is not a significant cause in this methodology. Furthermore, the relatively low VIF values compared with the conventional cutoff of 10 confirm the absence of multicollinearity, validating the robustness of the latent determinants for further assessment of the equation.

4.2 Empirical Analysis

Table 4 CSD & HS tests

Variables	Pesaran CD (stat)	Pesaran CD (p-value)
LnREC	3.79	0.000***
LnFD	20.98	0.000***
LnIQ	2.85	0.0781*
LnEI	28.26	0.000***
LnTO	12.4	0.000***
LnTI	16.32	0.000***
SH	Test result	
Δ	9.211***	
Δ̂_adjusted	11.425***	

Note. ***, **, and * denote 1%, 5%, and 10% statistical significance level respectively.

Table 4 summarizes the results of the CSD and HS tests for the explanatory and explained variables, respectively. The CSD test indicates significant cross-sectional dependence for most variables, whereas the HS test reveals diverse slopes in the dataset. The outcomes also suggest that the relationships between the dependent and independent variables in the model vary.

Table 5 Panel Unit Root test Results

Variables	LLC With Trend (t-stat)	LLC with Trend (p-value)	IPS with Trend (t-stat)	IPS with Trend (p-value)
LnREC	-2.4174	0.0078***	-5.0479	0.000***
LnFD	-3.6869	0.0001***	-1.8359	0.0332**
LnIQ	-1.9151	0.0277**	-6.9806	0.000***
LnEI	-4.1436	0.000***	-2.7489	0.003***
LnTO	-2.5684	0.0051***	5.3215	0.000***
LnTI	-2.9654	0.0015***	4.0904	0.000***

Note: Table 5 presents the panel unit root test outcomes employing Levin-Lin-Chu (LLC) and Im-Pesaran-Shin (IPS) unit root test methods. H_0 indicates panels containing unit roots. ***, **, and * represent 1%, 5%, and 10% statistical significance level respectively.

Next, we assessed the stationarity properties of the variables after confirming CSD and HS. Table 5 presents the results of the panel unit root tests. The results reveal that the determinants are stationary and that the panels do not suffer from unit roots.

Table 6 Pedroni panel cointegration Test

Test	Statistic	P-value
Modified Phillips-Perron t	2.719***	0.0033
Phillips-Perron t	(-2.9106)***	0.0018
Augmented Dickey-Fuller t	(-2.2811)**	0.0113

Note: ***, and ** indicate that the null hypothesis is rejected at the 1% and 5% significance levels, respectively.

Table 6 presents the results of the Pedroni cointegration test. The test statistics indicate significant cointegration at the 1% and 5% levels, respectively. In addition, this statistic implies that the variables share a long-term relationship with the dataset. Ultimately, there is evidence of cointegration among the variables under investigation.

Table 7 Granger Causality Test

Null Hypothesis	Z- Statistic	Z- Tilde Statistic	Granger Causality
LnFD → LnREC	15.5256***	11.8769***	Yes
LnIQ → LnREC	8.3647***	6.2726***	Yes
LnEI → LnREC	15.3131***	11.7106***	Yes
LnTO → LnREC	11.5454***	8.7618***	Yes
LnTI → LnREC	10.6134***	8.0325***	Yes

Note: ***, **, and * denote 1%, 5%, and 10% statistical significance levels, respectively.

Next, the Granger causality results presented in Table 7 reveal that causality flows from LnFD, LnIQ, LnEI, LnTO, and LnTI to REC. Table 7 further confirms that the model does not suffer from endogeneity, suggesting that the historical values of the independent variables can effectively indicate future changes in REC for the APEC region. Specifically, the significant causality between FD and REC highlights the critical roles of financial infrastructure, capital access, and investment in facilitating the transition to cleaner energy sources. Although Granger causality does not fully explain the underlying mechanisms, the results indicate that financial markets and institutions are essential drivers of renewable energy adoption, providing the necessary capital for green technology and infrastructure. This causal direction underscores the importance of FD in fostering the development of the renewable energy sector and building a more sustainable energy future.

Table 8 Hausman test

Test statistic	REC
Hausman test	23.12***(0.001)

Note. *** denote 1% statistical significance level.

Finally, the Hausman test in Table 8 strongly rejects the null hypothesis of no systematic difference between the FE and RE models. This significant result indicates that the individual effects are correlated with the independent variables, rejecting the assumptions of the RE model. Therefore, the FE model is more suitable for this analysis because it more accurately accounts for individual-specific effects.

4.2.1 Multivariate Analysis

Tabel 9 Regression Test

Independent Variables	Quantile Estimated Coefficients				Fixed Effects
	Q.25	Q.50	Q.75	Q.95	Coeffieicents
LnFD	-0.6694	-0.5386*	-0.4162	-0.2921	-0.5466***
	(-1.27)	(-1.8)	(-1.47)	(-0.58)	(-3.41)
LnIQ	-0.0148	0.0591	-0.1005	-0.1425	-0.0564
	(-0.03)	(-0.2)	(-0.35)	(-0.29)	(-0.38)
LnEI	0.7366	-0.5411	-0.3582	-0.1725	-0.5531**
	(-0.75)	(-0.98)	(-0.68)	(-0.19)	(-2.3)
LnTO	0.3012	0.1965	0.0986	-0.0008	0.2029
	0.55	0.63	0.34	0.18	1.5
LnTI	-0.3477*	-0.3345***	-0.3223***	-0.3098*	-0.3353***
	(-1.8)	(-3.06)	(-3.110)	(-1.73)	(-5.19)
					23.12***(0.001)

CD-test (P-value)	-	-	-	-	0.3826
Time Dummy	√	√	√	√	√
Country Dummy	√	√	√	√	√

Note: Table 6 exhibits FD impact on REC. The 25th, 50th, 75th, and 95th percentiles of LnREC are denoted. Z and t values are reported in parentheses for the quantile and fixed-effect regression approaches, respectively. ***, **, and * show the 1%, 5%, and 10% significance level respectively.

The outcomes from multiple panel estimation techniques are presented in Table 9, and the estimated coefficients for LnFD are negative across all quantiles: -0.6694 at Q.25 to -0.2921 at Q.95. The median quantile (Q.50, α = -0.5386) depicts that FD is statistically significant, indicating an inverse relationship between FD and REC. This outcome denotes that the REC tends to decrease as FD increases. The negative, statistically significant coefficient of LnFD (α = -0.5466) from the FE model also suggests an inverse relationship between FD and REC. The empirical results provide strong support for H1. This pattern suggests a misalignment in FD priorities, in which increased financial growth in middle-and high-income countries may not promote renewable energy adoption but rather rely on conventional energy sources. Institutional biases, entrenched energy systems, and risk-return preferences may drive financial flows toward traditional energy, diverting investments from renewable alternatives in APEC nations. These results are consistent with those of Al-Zubairi et al. (2025) and Osman et al. (2025), who found a negative association between FD and REC. However, these findings disagree with those of Achuo et al. (2025) and Rehman et al. (2025), who stated that FD increases REC in emerging economies.

Moreover, the coefficients for LnIQ were found to be negatively insignificant in both the FE model and across all quantiles. This outcome suggests that the impact of IQ on REC in the APEC region is either indirect or weak, implying that IQ alone does not play a pivotal role in the adoption of renewable energy in these economies. This outcome may be attributed to heterogeneous institutional structures across regions, where developed economies with high IQ may already have strong policy frameworks, while developing economies with weaker institutions may struggle to implement effective policies. For instance, Japan and South Korea have high IQ for the successful implementation of energy-efficient technologies and policies. In contrast, economies such as Indonesia and the Philippines have weaker institutional frameworks that may hinder the effective deployment of such policies for developing and deploying renewable energy technologies. Many APEC countries face challenges in improving governance mechanisms, such as policy enforcement, energy infrastructure, government subsidies, and international collaboration, along with global energy markets or international climate agreements. These outcomes align with the studies of Lao & Luo (2025) for South and East Asian economies and Van et al. (2025) for 116 countries, who indicated that weak institutional quality reduces renewable energy consumption. However, this outcome diverges from that of Rafiq et al. (2025), who reported that institutional quality promotes renewable energy consumption.

Furthermore, the analysis of energy intensity (EI) in APEC economies reveals a negative insignificant impact on REC across all quantiles, but a negative significant (α = - 0.5531) in the fixed effects model, suggesting that higher EI leads to weaker REC. This may be because of a heavy reliance on energy-intensive industries, such as Japan and China, which are part of the APEC region, and the challenges associated with transitioning to renewable energy without substantial infrastructure and technological advancements. These findings are consistent with those of Tian et al. (2025) for APEC and Degirmenci et al. (2025) for G7 countries, who revealed that economies with high energy intensity often rely on non-renewable energy sources. However, these results contrast with those of Feng et al. (2023), who stated that lower energy intensity enables more efficient renewable energy consumption.

Additionally, the coefficients of trade openness (TO) are positive but statistically insignificant at all quantiles and in the FE model. These results suggest that TO may improve access to renewable technologies; however, it does not directly lead to increased REC because of effective government policies, regulatory frameworks, economic incentives, and green technological infrastructure. These results further suggest that TO could increase EI due to the absence of stringent environmental policies. These outcomes are aligned with those of Ben Jebli & Ben Youssef (2015) for North Africa and Lin et al. (2016) for China, who reported that TO alone may not improve REC. Contrary to these findings, Jóźwik et al. (2025) argue that TO increases REC.

Finally, the findings in Table 9 indicate a significant but contrasting effect of LnTI on REC. This finding reflects that although TI has the potential to improve energy efficiency, it does not always directly contribute to the adoption of renewable energy within the APEC region. This outcome can be attributed to the heavy reliance on fossil fuels and slow policy adaptation for transitioning to green technologies. These findings are consistent with those of Popp (2019) and Adebayo et al. (2025) for G7 countries, who stated that TI hinders REC. However, the outcomes diverge from Alfalih (2025), who argued that TI has a beneficial effect on REC.

4.2.2 Further Investigation: Does IQ have a moderating role?

Table 10 The moderating effect of IQ on FD-REC nexus

Independent Variables	Upper Income	Upper Middle Income	APEC countries
LnFD	-1.0945***	-0.5766*	-1.6838***
	(-2.87)	(-1.49)	(-6.75)
LnIQR	-1.6539	1.1284	-3.3914***
	(-1.3)	(-1.42)	(-5.49)
LnFD*LnIQ	0.7899***	0.1242*	0.7815**
	3.3	0.54	4.91
Control Variables	√	√	√
Time Dummy	√	√	√
Country Dummy	√	√	√

Note: Table 10 exhibits the interaction effect of FD and IQ on REC when employing the FE approach. t values are exhibited in the parentheses for quantile and FE regression methods accordingly. ***, **, and * shows the significance value of 1%, 5%, and 10% respectively.

Table 10 reports the moderating role of IQ in the relationship between FD and REC in APEC economies. The results reveal that the interaction term LnFD*LnIQ exhibits a significant positive coefficient in APEC economies (3.3914, ***), indicating that institutional frameworks mitigate the negative effect of FD on REC. These results strongly support H2. This relationship is particularly relevant in the context of the APEC, where FD has often been associated with resource misallocation that prioritizes fossil fuel investments over clean energy alternatives. The negative coefficients for FD on REC in upper-income countries (-1.0945, *), upper-middle-income economies (-0.5766, *), and the broader APEC region (-1.6838, ***) indicate the risk of capital flowing into traditional, high-emission sectors. This trend can be attributed to the lack of policy alignment between financial growth and sustainable energy objectives and institutional weaknesses that fail to direct financial resources toward renewable energy technologies. However, the positive moderating effect of IQ (0.7899, ***) in APEC economies emphasizes the role of strong governance in shaping financial markets that prioritize green finance and sustainable investments. In economies with high IQ, such as those with stringent regulatory frameworks, enhanced transparency, and political stability, FD can drive innovation in renewable energy and facilitate the achievement of long-term environmental targets. These findings are in line with those of Chen et al. (2023), who emphasize the importance of IQ in shaping financial flows toward renewable energy projects, and Shafiullah et al. (2021) who highlight how effective institutional frameworks enhance the efficiency of financial systems in supporting clean energy. Conversely, Athari (2024) argued that technological advancements play a more substantial role in the renewable energy-financial development nexus. Thus, APEC economies can unlock the full potential of financial growth to support the transition toward a sustainable energy future by strengthening the institutional framework.

4.3 Robustness Result

Table 11
Robustness Test

	Replace the explanatory variable						Change the measurement method				
Independent Variables	Quantile Estimated Coefficients				Fixed Effects	GMM_SYS	MMQR Estimated Coefficients				Driscoll-Kraay
	Q.25	Q.50	Q.75	Q.95	Coeffieicents	Coeffieicents	Q.25	Q.50	Q.75	Q.95	Coeffieicents
LnFD	(-1.5449)***	(-1.163)	(-0.8424)	(-0.5076)	(-1.1858)***	(-0.468)**	(-0.6693)	(-0.5386)*	(-0.4162)*	(-0.2921)	(-0.5466)***
	(-3.32)	(-0.73)	(-0.30)	(-0.12)	(-5.01)	(-0.48)	(-1.33)	(-1.81)	(-1.69)	(-0.72)	(-3.11)
LnIQ	0.09629	0.0534	0.0174	(-0.0202)	(-0.0559)	-3.3837	(-0.0148)	(-0.0590)	(-0.1005)	(-0.1425)	(-0.0564)
	0.31	0.05	0.01	(-0.01)	0.39	(-2.18)	(-0.03)	(-0.20)	(-0.40)	(-0.36)	(-0.48)
LnEI	(-0.8875)*	(-0.682)	(-0.5095)	(-0.3293)	(-0.6942)**	(-0.1178)	(-0.7366)	(-0.5411)	(-0.3582)	(-0.1725)	(-0.5531)
	(-1.69)	(-0.38)	(-0.16)	(-0.07)	(-3.00)	(-0.11)	(-0.79)	(-0.98)	(-0.79)	(-0.23)	(-1.95)*
LnTO	0.6257*	0.4632	0.3267	0.1843	0.4729**	1.5244**	0.3012	0.1965	0.0986	(-0.0008)	0.2029
	1.77	0.38	0.15	0.06	3.16	2.23	0.58	0.63	0.39	(-0.00)	0.72
LnTI	(-0.3121)***	(-0.3071)	(-0.3029)	(-0.2986)	(-0.3074)***	(-0.1350)*	(-0.3476)*	(-0.3345)***	(-0.3223)***	(-0.3098)**	(-0.3353)
	-2.79	(-0.8)	(-0.45)	(-0.30)	(-4.91)	-0.33	(-1.88)	(-3.06)	(-3.57)	(-2.13)	(-6.34)***
CD-test (P-value)	-	-	-	-	0.4547	-	-	-	-	-	-
AR(1)	-	-	-	-	-	0.155	-	-	-	-	-
AR(2)	-	-	-	-	-	0.259	-	-	-	-	-
Hansen - Test	-	-	-	-	-	0.335	-	-	-	-	-
FC Dummy	√	√	√	√	√	√	-	-	-	-	-
Time Dummy	√	√	√	√	√	√	√	√	√	√	√
Country Dummy	√	√	√	√	√	√	√	√	√	√	√

Note: Table 11 presents the robustness test results for replacing the explanatory variable (FD1) and changing the methodology. The independent variable is replaced with quantile regression, Fixed Effects, and GMM_SYS models, while the methodology is altered to MMQR and Driscoll–Kraay. ***, **, and * shows the significance level of 1%, 5%, and 10% respectively.

To ensure the robustness of the results, multiple methods were used for robustness checks.

(1) Replacement of explanatory variables. This study replaces the measurement of explanatory variables by using stocks traded % of GDP from WDI to represent FD and conducts a regression analysis on the model. The results are presented in columns (1) to (6) in Table 11. The coefficient for LnFD2 in column (1) is significant at the 1% level, confirming a negative relationship between financial development and REC. Additionally, the fixed effects (FE) and SYS-GMM regression results presented in columns (5) and (6) of Table 11 further reinforce the negative relationship between FD and REC. This confirms that FD often reinforces fossil fuel biases, hindering REC growth in APEC economies. Furthermore, SYS-GMM addresses endogeneity issues, ensuring that the estimated effects are not driven by reverse causality. The results in column (6) of Table 11 indicate that the negative coefficient for LnFD (-0.468) remains robust even after accounting for endogeneity. The cross-sectional dependence test also reveals significant interdependence among APEC economies, justifying the use of SYS-GMM to address cross-country correlations. Finally, diagnostic tests confirm the validity of the model with no serial correlation (AR (1) p-value = 0.002) and no over-identification issues (Hansen p-value = 0.891). This strengthens the argument that FD is a key factor limiting REC, even after controlling for potential reverse causality.

(2) Replacement of econometric methods. To further enhance the robustness of our results, we performed regressions using the method of moment quantile regression (MMQR) and Driscoll–Kraay standard errors. The results are presented in columns (7) – (11) of Table 11. In the MMQR regression, LnFD is negative across all quantiles: -0.6693 at Q.25 to -0.2921 at Q.95. The median quantile (Q.50, α = -0.5386) and (Q.50, α = -0.4162) depict that FD is statistically significant, indicating an inverse relationship between FD and REC. The robustness of these results is reinforced by the Driscoll–Kraay standard errors, which address potential autocorrelation and heteroscedasticity, further confirming the negative impact of FD and the positive influence of technological integration. Therefore, these robustness results fully demonstrate that the finding that financial development hinders the energy transition is robust.

5. Conclusion and Policy Implications

This study investigates the relationship between FD and REC, focusing on the moderating role of IQ in APEC economies from 2002 to 2021. The study applies several econometric methods, including FE and panel quantile regression, and robustness checks, including the financial development proxy, SYS-GMM, MMQR, and Driscoll–Kraay standard error. The empirical findings reveal a negative association between FD and REC, especially at the median quantile, suggesting that FD in these economies is disproportionately directed toward traditional rather than renewable energy sectors. Notably, the analysis demonstrates that the negative impact of FD on REC is moderated by IQ, where higher IQ alleviates the adverse effects of FD, thereby fostering a more conducive environment for renewable energy investments. In addition, the study finds that EI is inversely related to REC, implying that economies with high energy consumption relative to GDP face significant barriers to transitioning to renewable energy without substantial technological advancements and infrastructure improvements. Although TO appears to have a limited direct impact on REC, it plays an essential role in facilitating the transfer of renewable energy technologies, particularly when supported by targeted policy interventions. Surprisingly, TI is found to have a negative effect on REC, suggesting a misalignment between innovation efforts and the region's renewable energy objectives.

5.1 Policy Suggestions

The conclusion suggests several policy recommendations for enhancing REC in APEC economies. First, the negative effect of FD on REC requires the reallocation of financial resources. Policymakers should consider creating a financial environment that encourages investment in renewable energy, such as through green bonds, subsidies, or tax incentives for renewable energy projects. This could support SDGs

7 (Affordable and Clean Energy) and 13 (Climate Action) by aligning financial resources with environmental goals. Second, improving IQ is critical for reducing the negative effects of FD on renewable energy adoption. Stronger institutions characterized by transparency, regulatory efficiency, and accountability may facilitate effective policy implementation and attract investments in renewable energy. Policymakers in APEC should focus on improving governance and institutional frameworks that support renewable energy initiatives, ensuring that financial resources are directed towards sustainable energy projects. Third, energy-intensive economies in the APEC require targeted policies to reduce their reliance on traditional energy sources. The negative relationship between EI and REC suggests that investing in energy efficiency is essential. Governments should encourage the adoption of energy-efficient technologies and reduce the carbon footprint of industrial processes to help economies shift towards cleaner energy sources. Fourth, TO should be leveraged to enhance the import and adoption of renewable energy technologies. Although TO alone does not directly drive REC, it facilitates access to advanced renewable energy technologies. APEC economies should reduce trade barriers, remove tariffs on renewable energy technologies, and encourage the adoption of green technologies through incentives and policy frameworks to achieve this goal. Finally, TI policies must be realigned to focus on renewable energy objectives. Despite its association with energy efficiency, TI in APEC economies currently negatively impacts REC. Governments should foster innovation that directly contributes to renewable energy technologies and promote research and development in clean energy sectors.

5.2 Limitations and Future Direction

This study has several limitations that should be addressed in future research. First, the analysis is based on data from only 12 APEC countries due to data unavailability. A broader sample, potentially including more APEC economies or countries from other regions, would enhance the generalizability of these findings to other countries. Moreover, this study employs a specific set of control variables, and future studies should consider incorporating other important variables, such as carbon pricing, environmental regulations, and political risk, to provide a more comprehensive understanding of renewable energy consumption dynamics.

Additionally, while quantile regression offers valuable insights into the relationships across different distribution points, it does not account for potential nonlinearities or threshold effects that may exist between variables. Future research should incorporate these considerations for a more nuanced analysis. The reliance on FE and RE models may also overlook the dynamic nature of the data. Advanced methodologies, such as structural equation modeling (SEM), could offer deeper insights into these relationships. The study also presents IQ as a moderating factor but does not explore its multidimensional aspects, such as corruption, governance transparency, and political stability. Finally, external factors, including the multidimensional aspects of IQ, geopolitical risks, global energy prices, and technological advancements, were not considered and could provide additional context for understanding renewable energy consumption dynamics. Future research should address these areas to enrich the current findings.

Acknowledgements: The authors would like to thank the anonymous institution for supporting this research.

Author Contributions: M.A. S. C. and M.A. originated and intended the research, M.A., and M.H. completed the literature survey on the topic, M.A. S. C., and M.A. accomplished the statistical facts, M.H. approved funding acquisition, M.A. S. C., and M.A. finalized and prepared the draft. Corresponding author: M.A.. All authors contributed to the writing and approved the final manuscript.

Declaration of Generative AI and AI-assisted technologies in the writing process: During the preparation of this work, the authors used OpenAI's ChatGPT for assistance with language generation during the writing process. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Competing Interests: None of the authors have a conflict of interest to disclose.

Ethical Approval: This article does not contain any studies with human participants performed by any of the authors.

Informed Consent: This article does not contain any studies with human participants performed by any of the authors.

Data Availability: Data is provided within manuscript or supplementary information files.

References

Abbasi, F., & Riaz, K. (2016). CO2 emissions and financial development in an emerging economy: an augmented VAR approach. *Energy Policy*, *90*, 102–114.

Acheampong, A. O., Dzator, J., & Savage, D. A. (2021). Renewable energy, CO2 emissions and economic growth in sub-Saharan Africa: does institutional quality matter? *Journal of Policy Modeling*, *43*(5), 1070–1093.

Achuo, E., Kakeu, P., & Asongu, S. (2025). Financial development, human capital and energy transition: a global comparative analysis. *International Journal of Energy Sector Management*, *19*(1), 59–80. https://doi.org/10.1108/IJESM-11-2023-0004

Adebayo, T. S., Staniewski, M. W., & Uzun, B. (2025). Policy implications of technological innovations, domestic material consumption and renewable energy consumption for achieving sustainable development goals in G7 economies. *International Journal of Sustainable Development and World Ecology*, *32*(4), 465–484. https://doi.org/10.1080/13504509.2025.2487793

Ahmad, M., Peng, T., Awan, A., & Ahmed, Z. (2023). Policy framework considering resource curse, renewable energy transition, and institutional issues: Fostering sustainable development and sustainable natural resource consumption practices. *Resources Policy*, *86*, 104173. https://doi.org/10.1016/j.resourpol.2023.104173

Alfalih, A. A. (2025). Assessing the impact of green technological innovation and human capabilities on renewable energy supply: Evidence from the top-greenest countries. *Sustainable Energy Technologies and Assessments*, *73*, 104152. https://doi.org/10.1016/j.seta.2024.104152

Alshagri, R., Alsabhan, T. H., & Binsuwadan, J. (2024). Investigating the Role of Financial Development in Encouraging the Transition to Renewable Energy: A Fractional Response Model Approach. *Sustainability (Switzerland)*, *16*(18). https://doi.org/10.3390/su16188153

Al-Zubairi, A., AL-Akheli, A., & ELfarra, B. (2025). The impact of financial development, renewable energy and political stability on carbon emissions: sustainable development prospective for arab economies. *Environment, Development and Sustainability*, *27*(7), 15251–15273. https://doi.org/10.1007/s10668-024-04703-5

Amenta, E., & Ramsey, K. M. (2010). Institutional Theory. In *Handbooks of Sociology and Social Research* (pp. 15–39). Springer. https://doi.org/10.1007/978-0-387-68930-2_2

Apergis, N., & Pinar, M. (2021). The role of party polarization in renewable energy consumption: Fresh evidence across the EU countries. *Energy Policy*, *157*, 112518. https://doi.org/10.1016/j.enpol.2021.112518

Arminen, H., & Menegaki, A. N. (2019). Corruption, climate and the energy-environment-growth nexus. *Energy Economics*, *80*, 621–634.

Athari, S. A. (2024). Global economic policy uncertainty and renewable energy demand: Does environmental policy stringency matter? Evidence from OECD economies. *Journal of Cleaner Production*, *450*, 141865. https://doi.org/10.1016/j.jclepro.2024.141865

Baltagi, B. (2005). Econometric analysis of panel data. In *Vasa* (Vol. 4). Springer.

Basheer, M. F., Sabir, S. A., & Hassan, S. G. (2024). Financial development, globalization, energy consumption, and environmental quality: Does control of corruption matter in South Asian countries? *Economic Change and Restructuring*, *57*(3), 112. https://doi.org/10.1007/s10644-024-09699-6

Behera, B., Behera, P., & Sethi, N. (2024). Decoupling the role of renewable energy, green finance and political stability in achieving the sustainable development goal 13: Empirical insight from emerging economies. *Sustainable Development*, *32*(1), 119–137. https://doi.org/10.1002/sd.2657

Behera, B., Sucharita, S., Patra, B., & Sethi, N. (2024). A blend of renewable and non-renewable energy consumption on economic growth of India: The role of disaggregate energy sources. *Environmental Science and Pollution Research International*, *31*(3), 3902–3916. https://doi.org/10.1007/s11356-023-31372-0

Ben Jebli, M., & Ben Youssef, S. (2015). Economic growth, combustible renewables and waste consumption, and CO2 emissions in North Africa. *Environmental Science and Pollution Research*, *22*(20), 16022–16030. https://doi.org/10.1007/s11356-015-4792-0

Chen, Y., Chen, Y., Zhang, L., & Li, Z. (2023). Revealing the role of renewable energy consumption and digitalization in energy-related greenhouse gas emissions—Evidence from the G7. *Frontiers in Energy Research*, *11*, 1197030. https://doi.org/10.3389/fenrg.2023.1197030

Clémençon, R. (2024). How important are international climate negotiations? In *Routledge Handbook of Climate Change and Society* (pp. 411–425). Routledge. https://doi.org/10.4324/9781003291206-34

Danish, Saud, S., Baloch, M. A., & Lodhi, R. N. (2018). The nexus between energy consumption and financial development: estimating the role of globalization in Next-11 countries. *Environmental Science and Pollution Research*, *25*(19), 18651–18661. https://doi.org/10.1007/s11356-018-2069-0

Degirmenci, T., Sofuoglu, E., Aydin, M., & Adebayo, T. S. (2025). The role of energy intensity, green energy transition, and environmental policy stringency on environmental sustainability in G7 countries. *Clean Technologies and Environmental Policy*, *27*(7), 2981–2993. https://doi.org/10.1007/s10098-024-02968-y

Driscoll, J. C., & Kraay, A. C. (1998). Consistent covariance matrix estimation with spatially dependent panel data. *Review of Economics and Statistics*, *80*(4), 549–559. https://doi.org/10.1162/003465398557825

Farhani, S., & Solarin, S. A. (2017). Financial development and energy demand in the United States: New evidence from combined cointegration and asymmetric causality tests. *Energy*, *134*, 1029–1037. https://doi.org/10.1016/j.energy.2017.06.121

Feng, Y., Zhang, J., Geng, Y., Jin, S., Zhu, Z., & Liang, Z. (2023). Explaining and modeling the reduction effect of low-carbon energy transition on energy intensity: Empirical evidence from global data. *Energy*, *281*, 128276. https://doi.org/10.1016/j.energy.2023.128276

Gani, A. (2021). Sustainability of energy assets and corruption in the developing countries. *Sustainable Production and Consumption*, *26*, 741–751. https://doi.org/10.1016/j.spc.2020.12.023

Geels, F. W. (2002). Technological transitions as evolutionary reconfiguration processes: A multi-level perspective and a case-study. *Research Policy*, *31*(8–9), 1257–1274. https://doi.org/10.1016/S0048-7333(02)00062-8

Geels, F. W., Sovacool, B. K., Schwanen, T., & Sorrell, S. (2017). The socio-technical dynamics of low-carbon transitions. *Joule*, *1*(3), 463–479.

Granger, C. W. J. (1969). Investigating Causal Relations by Econometric Models and Cross-spectral Methods. *Econometrica*, *37*(3), 424. https://doi.org/10.2307/1912791

Hasanov, F. J., Khan, Z., Hussain, M., & Tufail, M. (2021). Theoretical Framework for the Carbon Emissions Effects of Technological Progress and Renewable Energy Consumption. *Sustainable Development*, *29*(5), 810–822. https://doi.org/10.1002/sd.2175

Hashem Pesaran, M., & Yamagata, T. (2008). Testing slope homogeneity in large panels. *Journal of Econometrics*, *142*(1), 50–93. https://doi.org/10.1016/j.jeconom.2007.05.010

Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica: Journal of the Econometric Society*, 1251–1271.

Hsiao, C. (2022). *Analysis of panel data* (Issue 64). Cambridge university press.

Huck, W. (2023). Transforming our world: the 2030 Agenda for Sustainable Development. *Sustainable Development Goals*, *1*, 41. https://doi.org/10.5040/9781509934058.0025

Hussain, M., & Dogan, E. (2021). The role of institutional quality and environment-related technologies in environmental degradation for BRICS. *Journal of Cleaner Production*, *304*, 127059. https://doi.org/10.1016/j.jclepro.2021.127059

Im, K. S., Pesaran, M. H., & Shin, Y. (2023). Reprint of: Testing for unit roots in heterogeneous panels. *Journal of Econometrics*, *234*(1), 56–69. https://doi.org/10.1016/j.jeconom.2023.03.002

Islam, M. M., Ali, M. I., Ceh, B., Singh, S., Khan, M. K., & Dagar, V. (2022). Renewable and non-renewable energy consumption driven sustainable development in ASEAN countries: do financial development and institutional quality matter? *Environmental Science and Pollution Research*, *29*(23), 34231–34247. https://doi.org/10.1007/s11356-021-18488-x

Jóźwik, B., Sarıgül, S. S., Topcu, B. A., Çetin, M., & Doğan, M. (2025). Trade Openness, Economic Growth, Capital, and Financial Globalization: Unveiling Their Impact on Renewable Energy Consumption. *Energies*, *18*(5), 1244. https://doi.org/10.3390/en18051244

Junejo, S., Shaikh, N. A., & Yang, L. (2026). Trade Liberalization and the Elusive Growth Dividend in Less Developed Countries: A Panel Data Analysis with Cross-Sectional Dependence. In *Integrating Big Data and IoT for Enhanced Decision-Making Systems in Business: Volume 1* (pp. 221–248). Springer. https://doi.org/10.1007/978-3-031-97609-4_20

Khan, Z., Hussain, M., Shahbaz, M., Yang, S., & Jiao, Z. (2020). Natural resource abundance, technological innovation, and human capital nexus with financial development: A case study of China. *Resources Policy*, *65*, 101585. https://doi.org/10.1016/j.resourpol.2020.101585

Khatib, S. F. A. (2025). An assessment of methods to deal with endogeneity in corporate governance and reporting research. *Corporate Governance: The International Journal of Business in Society*, *25*(3), 606–630.

Lao, G. C., & Luo, G. Q. (2025). Linkages between industry, agriculture, growth, renewable energy and the environment and the role of institutional quality, evidence from selected South and East Asian economies. *Energy Exploration and Exploitation*, *43*(1), 354–377. https://doi.org/10.1177/01445987241268030

Levin, A., Lin, C. F., & Chu, C. S. J. (2002). Unit root tests in panel data: Asymptotic and finite-sample properties. *Journal of Econometrics*, *108*(1), 1–24. https://doi.org/10.1016/S0304-4076(01)00098-7

Li, B. (2023). The role of financial markets in the energy transition: an analysis of investment trends and opportunities in renewable energy and clean technology. *Environmental Science and Pollution Research*, *30*(43), 97948–97964. https://doi.org/10.1007/s11356-023-29014-6

Lin, B., Omoju, O. E., & Okonkwo, J. U. (2016). Factors influencing renewable electricity consumption in China. *Renewable and Sustainable Energy Reviews*, *55*, 687–696. https://doi.org/10.1016/j.rser.2015.11.003

Liu, W., Shen, Y., & Razzaq, A. (2023). How renewable energy investment, environmental regulations, and financial development derive renewable energy transition: Evidence from G7 countries. *Renewable Energy*, *206*, 1188–1197. https://doi.org/10.1016/j.renene.2023.02.017

Makhdum, M. S. A., Usman, M., Kousar, R., Cifuentes-Faura, J., Radulescu, M., & Balsalobre-Lorente, D. (2022). How Do Institutional Quality, Natural Resources, Renewable Energy, and Financial Development Reduce Ecological Footprint without Hindering Economic Growth Trajectory? Evidence from China. *Sustainability (Switzerland)*, *14*(21), 13910. https://doi.org/10.3390/su142113910

Mesagan, E. P., & Olunkwa, C. N. (2022). Heterogeneous analysis of energy consumption, financial development, and pollution in Africa: The relevance of regulatory quality. *Utilities Policy*, *74*, 101328. https://doi.org/10.1016/j.jup.2021.101328

Mustafa, M., Ghazali, F., & Yuniza, M. E. (2022). Governance interventions in providing affordable, reliable, sustainable, and modern energy in the ASEAN region (Goal 7). In *Good Governance and* *the Sustainable Development Goals in Southeast Asia* (pp. 84–97). Routledge. https://doi.org/10.4324/9781003230724-8

Mwanjilinji, E. E., Shakiru, T. H., & Huang, F.-M. (2025). The Role of Credit Risk in Shaping the Profitability of the Banking Sector in the World's Largest Economies: Evidence from Panel Quantile Regression. *SN Business \& Economics*, *5*(11), 1–30.

North, D. C. (2005). The Contribution of the New Institutional Economics to an Understanding of the Transition Problem. In *Wider Perspectives on Global Development* (pp. 1–15). Springer. https://doi.org/10.1057/9780230501850_1

Ofori, E. K., Hayford, I. S., Nyantakyi, G., Tergu, C. T., & Opoku-Mensah, E. (2023). Synerging Sustainable Development Goals—can clean energy (green) deliver UN-SDG geared towards socio-economic-environment objectives in emerging BRICS? *Environmental Science and Pollution Research*, *30*(43), 98470–98489. https://doi.org/10.1007/s11356-023-29209-x

Osman, A., Afjal, M., Alharthi, M., Elheddad, M., Djellouli, N., & He, Z. (2025). Green growth dynamics: unraveling the complex role of financial development and natural resources in shaping renewable energy in Sub-Saharan Africa. *Humanities and Social Sciences Communications*, *12*(1), 64. https://doi.org/10.1057/s41599-025-04364-3

Ouyang, Y., & Li, P. (2018). On the nexus of financial development, economic growth, and energy consumption in China: New perspective from a GMM panel VAR approach. *Energy Economics*, *71*, 238–252. https://doi.org/10.1016/j.eneco.2018.02.015

Pedroni, P. (2004). PANEL COINTEGRATION: ASYMPTOTIC AND FINITE SAMPLE PROPERTIES OF POOLED TIME SERIES TESTS WITH AN APPLICATION TO THE PPP HYPOTHESIS. *Econometric Theory*, *20*(03), 597–625. https://doi.org/10.1017/S0266466604203073

Pesaran, M. H. (2021). General diagnostic tests for cross-sectional dependence in panels. *Empirical Economics*, *60*(1), 13–50. https://doi.org/10.1007/s00181-020-01875-7

Polzin, F., Sanders, M., & Täube, F. (2017). A diverse and resilient financial system for investments in the energy transition. *Current Opinion in Environmental Sustainability*, *28*, 24–32. https://doi.org/10.1016/j.cosust.2017.07.004

Popp, D. (2019). Environmental policy and innovation: A decade of research. *International Review of Environmental and Resource Economics*, *13*(3–4), 265–337. https://doi.org/10.1561/101.00000111

Potrafke, N. (2010). Does government ideology influence deregulation of product markets? Empirical evidence from OECD countries. *Public Choice*, *143*(1), 135–155. https://doi.org/10.1007/s11127-009-9494-z

Rafiq, S., Paramati, S. R., Alam, M. S., Hafeez, K., & Shafiullah, M. (2025). Does institutional quality matter for renewable energy promotion in OECD economies? *International Journal of Finance and Economics*, *30*(1), 477–492. https://doi.org/10.1002/ijfe.2926

Rehman, A., Batool, Z., Ain, Q. U., & Ma, H. (2025). The renewable energy challenge in developing economies: An investigation of environmental taxation, financial development, and political stability. *Natural Resources Forum*, *49*(1), 699–724. https://doi.org/10.1111/1477-8947.12418

Riti, J. S., Shu, Y., & Kamah, M. (2021). Institutional quality and environmental sustainability: The role of freedom of press in most freedom of press countries. *Environmental Impact Assessment Review*, *91*, 106656. https://doi.org/10.1016/j.eiar.2021.106656

Saadaoui, H., & Chtourou, N. (2023). Do Institutional Quality, Financial Development, and Economic Growth Improve Renewable Energy Transition? Some Evidence from Tunisia. *Journal of the Knowledge Economy*, *14*(3), 2927–2958. https://doi.org/10.1007/s13132-022-00999-8

Saba, C. S., & Biyase, M. (2022). Determinants of renewable electricity development in Europe: Do Governance indicators and institutional quality matter? *Energy Reports*, *8*, 13914–13938. https://doi.org/10.1016/j.egyr.2022.09.184

Saliba, C., Farmanesh, P., & Athari, S. A. (2023). Does country risk impact the banking sectors' non-performing loans? Evidence from BRICS emerging economies. *Financial Innovation*, *9*(1), 86.

Shafiullah, M., Miah, M. D., Alam, M. S., & Atif, M. (2021). Does economic policy uncertainty affect renewable energy consumption? *Renewable Energy*, *179*, 1500–1521. https://doi.org/10.1016/j.renene.2021.07.092

Shahbaz, M., Shahzad, S. J. H., Ahmad, N., & Alam, S. (2016). Financial development and environmental quality: The way forward. *Energy Policy*, *98*, 353–364. https://doi.org/10.1016/j.enpol.2016.09.002

Shahbaz, M., Topcu, B. A., Sarıgül, S. S., & Vo, X. V. (2021). The effect of financial development on renewable energy demand: The case of developing countries. *Renewable Energy*, *178*, 1370–1380. https://doi.org/10.1016/j.renene.2021.06.121

Song, X., & Taamouti, A. (2021). Measuring Granger Causality in Quantiles. *Journal of Business and Economic Statistics*, *39*(4), 937–952. https://doi.org/10.1080/07350015.2020.1739531

Sun, G., Li, G., Dilanchiev, A., & Kazimova, A. (2023). Promotion of green financing: Role of renewable energy and energy transition in China. *Renewable Energy*, *210*, 769–775. https://doi.org/10.1016/j.renene.2023.04.044

Sun, Z., Zhang, X., & Gao, Y. (2023). The Impact of Financial Development on Renewable Energy Consumption: A Multidimensional Analysis Based on Global Panel Data. *International Journal of Environmental Research and Public Health*, *20*(4), 3124. https://doi.org/10.3390/ijerph20043124

Tian, G., Hua, X., Li, D., & Tian, J. (2025). How energy intensity and global energy dynamics shape renewable energy transition in APEC economies. *Geoscience Frontiers*, *16*(6), 102134. https://doi.org/10.1016/j.gsf.2025.102134

Troster, V. (2018). Testing for Granger-causality in quantiles. *Econometric Reviews*, *37*(8), 850–866. https://doi.org/10.1080/07474938.2016.1172400

Van, H. N., Quoc, H. N., & Le Quoc, D. (2025). Towards Sustainable Development: Drivers From Financial and Institutional Development. *Journal of Public Affairs*, *25*(3), e70073. https://doi.org/10.1002/pa.70073

Wang, E., Gozgor, G., Mahalik, M. K., Patel, G., & Hu, G. (2022). Effects of institutional quality and political risk on the renewable energy consumption in the OECD countries. *Resources Policy*, *79*, 103041. https://doi.org/10.1016/j.resourpol.2022.103041

Wang, J., Zhang, S., & Zhang, Q. (2021). The relationship of renewable energy consumption to financial development and economic growth in China. *Renewable Energy*, *170*, 897–904. https://doi.org/10.1016/j.renene.2021.02.038

Wang, Y., Hanif, N., Arshed, N., Hussain, M., Arshad, Z., & Usman, M. (2024). Synergizing social sustainability and technological innovation for carbon neutrality in OECD economies. *Sustainable Development*, *32*(6), 6614–6628. https://doi.org/10.1002/sd.3038

Wen, J., Hao, Y., Feng, G. F., & Chang, C. P. (2016). Does government ideology influence environmental performance? Evidence based on a new dataset. *Economic Systems*, *40*(2), 232–246. https://doi.org/10.1016/j.ecosys.2016.04.001

Xu, P., Hussain, M., Ye, C., Wang, J., Wang, C., Geng, J., Liu, Y., & Chen, J. (2022). Natural resources, economic policies, energy structure, and ecological footprints' nexus in emerging seven countries. *Resources Policy*, *77*, 102747. https://doi.org/10.1016/j.resourpol.2022.102747

Yadav, A., Bekun, F. V., Ozturk, I., Ferreira, P. J. S., & Karalinc, T. (2024). Unravelling the role of financial development in shaping renewable energy consumption patterns: Insights from BRICS countries. *Energy Strategy Reviews*, *54*, 101434. https://doi.org/10.1016/j.esr.2024.101434

Yue, S., Lu, R., Shen, Y., & Chen, H. (2019). How does financial development affect energy consumption? Evidence from 21 transitional countries. *Energy Policy*, *130*, 253–262. https://doi.org/10.1016/j.enpol.2019.03.029

Zhang, P., Li, Z., Ghardallou, W., Xin, Y., & Cao, J. (2023). Nexus of institutional quality and technological innovation on renewable energy development: Moderating role of green finance. *Renewable Energy*, *214*, 233–241. https://doi.org/10.1016/j.renene.2023.05.089

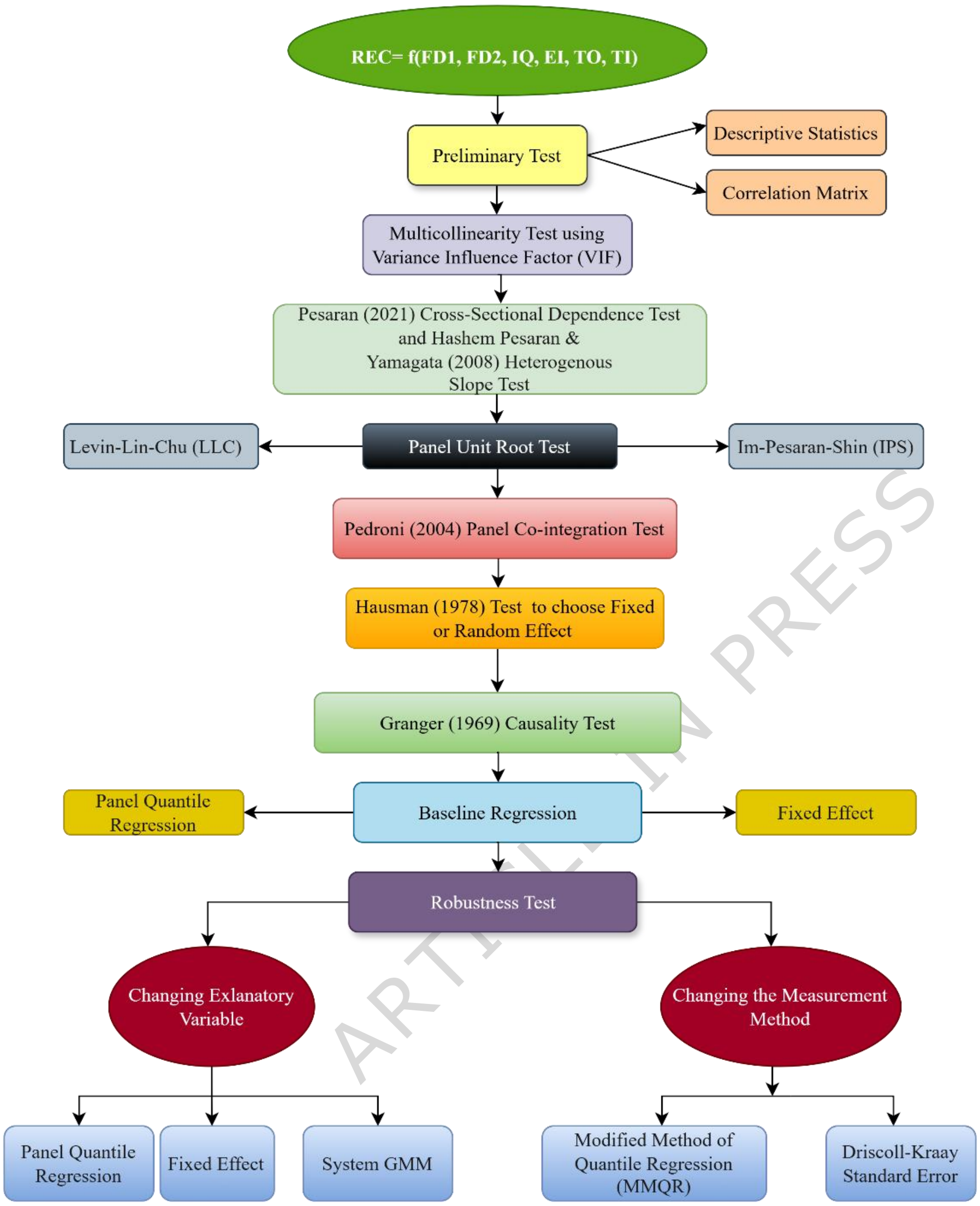

Figure 1: Research methodology flowchart